

AI Bio Researcher Intern Exercise

Predicting Cas9 Gene-Editor Activity

SUBHASH KURMANA - ED23B066

1. Initial understanding and problem framing

Before starting the implementation, I first spent time understanding the problem itself. I went through the research paper provided with the assignment to understand the biological setting, how Cas9 activity was measured using deep mutational scanning (DMS), and why the proposed approach combines pretrained protein representations with structure models. Several of the terms which were new to me, so I also looked up the relevant concepts and documentation separately. I wanted to understand what each component actually represented before deciding how to use it in the experiment.

After this initial study, I inspected the supplied assay and the wild-type Cas9 reference sequence. I treated the proposed architecture as a hypothesis to test rather than assuming that a more complex model would necessarily be better. The main question I then used to guide the work was: can a frozen pretrained sequence representation improve prediction of Cas9 activity on mutation positions that were not seen during training?

2. What the measurements support predicting

The supplied dataset contains 8,117 single Cas9 mutants. The target is DMS_score from a positive-selection assay, so I interpret it as an assay-specific activity/fitness phenotype rather than as a universal measure of gene-editing performance. The assay is influenced by the experimental setting and by factors such as expression and correct cleavage. Therefore, my evaluation is deliberately framed around predicting the supplied activity-related score within this dataset.

3. Experimental workflow

I followed the workflow below so that each modelling decision was made only after checking what the data and the previous experiment established.

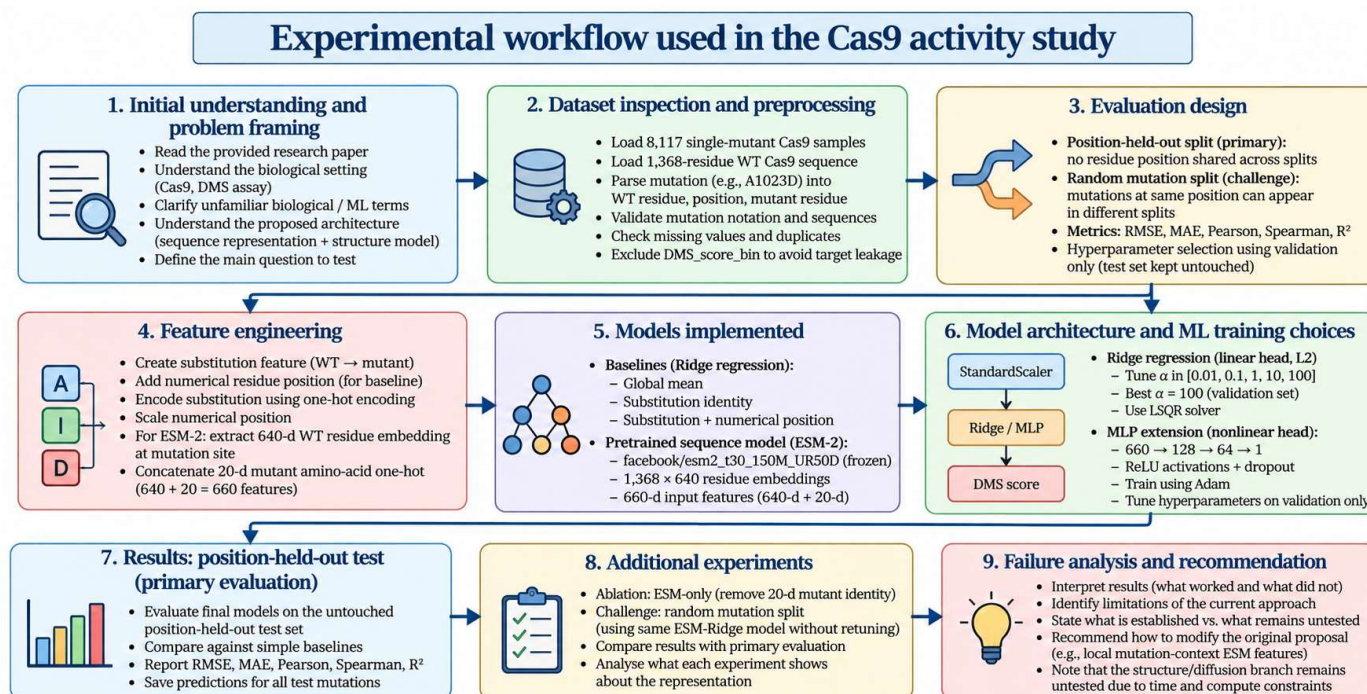


Figure 1. Experimental workflow used to evaluate the proposed Cas9 activity-prediction approach.

4. Evaluation design

I used two complementary splits. The primary split is position-held-out: all mutations at a residue position are kept in the same split, so no position is shared across train, validation and test. This was chosen because the main generalization claim I wanted to test was prediction at previously unseen positions. I also created a random mutation split as a separate challenge experiment; in this split, different mutations at the same position can occur in different splits, so it is a more interpolation-like setting.

I used RMSE and MAE to measure numerical prediction error, together with Pearson and Spearman correlation to check whether the predictions tracked experimental variation and rank order. R^2 was also reported. Hyperparameters were selected using validation data only, and the primary test set was held out until the final evaluation.

5. Data preprocessing and feature engineering

I first parsed each mutation such as A1023D into three explicit fields: wild-type residue, residue position, and mutant residue. I checked that the wild-type residue agreed with the 1,368-residue WT FASTA and that the provided mutated sequence was consistent with the mutation notation. I also checked missing values, duplicate mutations and duplicate sequences. The DMS_score_bin column was not used as a feature because it is derived from the target and would introduce target leakage.

Stage	Representation	Size	Why I used it
Classical baseline	A->D categorical + scaled position	categorical + 1	Tests substitution alone, then asks whether position adds signal.
Pretrained residue feature	WT residue embedding from ESM-2	640-d	Transfers sequence-derived protein representation.
Final ESM feature	640-d embedding + mutant one-hot	660-d	Explicit interface given to Ridge and MLP heads.

For the classical baselines, I used scikit-learn pipelines with OneHotEncoder for substitution identity and StandardScaler for numerical position. For ESM-2, I encoded residues 1-1022 and 347-1368 as two overlapping 1022 x 640 matrices, then assembled a 1368 x 640 residue matrix by taking the representation farther from a window boundary. This let me precompute one residue table instead of rerunning ESM-2 for every mutation.

6. Model architecture and ML training choices

- Encoder/interface: facebook/esm2_t30_150M_UR50D was frozen. After removing special tokens, last_hidden_state is [1, L, 640]; I selected the WT residue vector and concatenated the 20-d mutant one-hot vector to form x in R^{660} . This is fixed-feature transfer learning, not end-to-end fine-tuning.
- Linear head: StandardScaler -> Ridge with L2 regularization. I searched alpha in [0.01, 0.1, 1, 10, 100], selected 100 on validation data, and used the LSQR solver for stable numerical behavior.
- Nonlinear extension: StandardScaler -> MLP 660 -> 128 -> 64 -> 1 with ReLU and Adam. I tested alpha in [1e-4, 1e-3, 1e-2] and selected 1e-2 on validation data. Only the head was trained.

After model selection, I retrained on train + validation and evaluated once on the untouched test set, preventing test leakage into tuning.

7. Models I implemented

I first established simple baselines before introducing a pretrained model. This gives a reference for whether the added representation is actually useful rather than assuming that model complexity itself is an improvement.

- Global mean: predicts the mean training-set activity for every mutation.
- Substitution identity: represents the wild-type to mutant amino-acid change, e.g. A->D.
- Substitution + position: adds the numerical residue position to the substitution feature.

For the pretrained sequence experiment, I used the Hugging Face checkpoint facebook/esm2_t30_150M_UR50D. The ESM-2 encoder was frozen and used only as a feature extractor. Because Cas9 is 1,368 residues long and the model input window is limited, I used two overlapping sequence windows and combined them into a 1,368 x 640 residue-embedding matrix. For each mutation, the predictor received the 640-dimensional WT residue embedding at that position together with a 20-dimensional one-hot representation of the mutant amino acid, giving 660 input features.

- Ridge regression on the 660-dimensional representation.
- MLP extension: 660 -> 128 -> 64 -> 1, trained only after the linear predictor was established.

8. Primary results: position-held-out test

The primary results are shown below. I used the held-out test set only after selecting the model settings on validation data.

Model	RMSE	MAE	Pearson	Spearman	R^2
Global mean	0.5855	0.3969	-	-	~0
Substitution	0.5846	0.3943	0.0935	0.1396	0.0031
Substitution + Position	0.5855	0.3949	0.0864	0.1243	~0
ESM-2 + Ridge	0.6042	0.4310	0.1893	0.1791	-0.0648
ESM-2 + MLP	0.7218	0.5348	0.0704	0.0792	-0.5196

The substitution baseline achieved the lowest RMSE and MAE on the primary test set. The ESM-2 + Ridge model showed higher Pearson and Spearman correlations, which means it captured some signal in the experimental scores, but its overall numerical error was still larger. The MLP did not improve on the linear Ridge head and instead generalized worse on this split.

Interpretation: the tested frozen single-position ESM-2 interface did not improve prediction error over the simple substitution baseline for unseen residue positions.

9. Ablation and separate challenge experiment

I then removed the explicit 20-dimensional mutant identity and kept only the 640-dimensional ESM-2 residue representation. This ESM-only ablation gave RMSE 0.6016, MAE 0.4285, Pearson 0.1917, Spearman 0.1902 and R^2 -0.0559 on the same position-held-out test. Its similarity to the full ESM-Ridge result suggests that the explicit mutant one-hot feature made little difference in this particular interface.

As a separate challenge, I evaluated the selected ESM-Ridge model on the random mutation split without retuning it. It gave RMSE 0.6437, MAE 0.4325, Pearson 0.1774, Spearman 0.2025 and R^2 0.0067. The random-split substitution + position baseline had RMSE 0.6384. The ESM model therefore did not provide a clear improvement in this interpolation-like setting either.

10. Failure analysis and recommendation

The main limitation I identified is the representation interface itself. I used a single WT residue embedding at the mutation position, together with the mutant identity. This does not explicitly describe the local sequence context of the mutation, nor does it encode the full mutated sequence. A single-position representation can therefore miss interactions between the mutation and nearby residues. The nonlinear extension also did not recover this missing information in the current experiment.

Based on the evidence I obtained, I would modify the original proposal rather than pursue it unchanged. I would keep pretrained sequence representations as a candidate component, but I would first test a local mutation-context ESM representation and compare it against the same position-held-out baseline. The diffusion/structure branch should then be tested as a separate experiment before being integrated into the final architecture. I did not execute the structure/diffusion branch within the available time and compute budget, so I do not use the current results to claim that structural features are ineffective.

11. What I established, what remains open, and next step

Established in this study: (1) the assay can support prediction of its activity-related DMS score within the supplied experimental setting; (2) the position-held-out split provides a stricter test of generalization to unseen residue positions; (3) simple mutation features form a strong baseline for this task; and (4) the tested frozen ESM-2 single-position interface did not reduce RMSE or MAE relative to that baseline.

Not established: whether a better sequence interface, a structure representation, or a joint sequence-structure model would improve performance. In particular, I did not train or evaluate a diffusion-based structure branch, so that part of the original architectural proposal remains an open question.

My next experiment would therefore be a controlled sequence-context study: represent a local window around each mutation with ESM-2, include the mutant identity, and compare it against the existing substitution baseline under the same position-held-out split. Only if that representation shows consistent improvement would I move to an isolated structure/diffusion experiment and then consider a combined architecture.

12. Reproducibility and work record

- Environment: Python 3.13; CPU used for ESM-2 inference and downstream model training.
- Notebook order: 01_data_preprocessing_and_splitting.ipynb -> 02_baselines_and_protein_model.ipynb -> 03_esm2_activity_prediction.ipynb.
- Saved outputs include processed data, fixed split assignments, ESM-2 residue embeddings, held-out predictions, baseline results, ESM results, ablation results and challenge results.
- AI tools: ChatGPT was used for experiment planning, coding assistance, debugging and interpretation. I checked and ran the resulting code locally in the repository.
- Unfinished within the time limit: direct diffusion/structure-feature experiment and the local-context ESM extension.
- The assignment time limit was treated as a hard constraint; I stopped after completing the baseline, ESM-2 predictor, ablation and challenge evaluation needed to support a recommendation.

References

Spencer JM and Zhang X. Deep mutational scanning of *S. pyogenes* Cas9 reveals important functional domains. *Scientific Reports* 7, 16836 (2017).
Notin et al. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. Official ProteinGym benchmark repository.